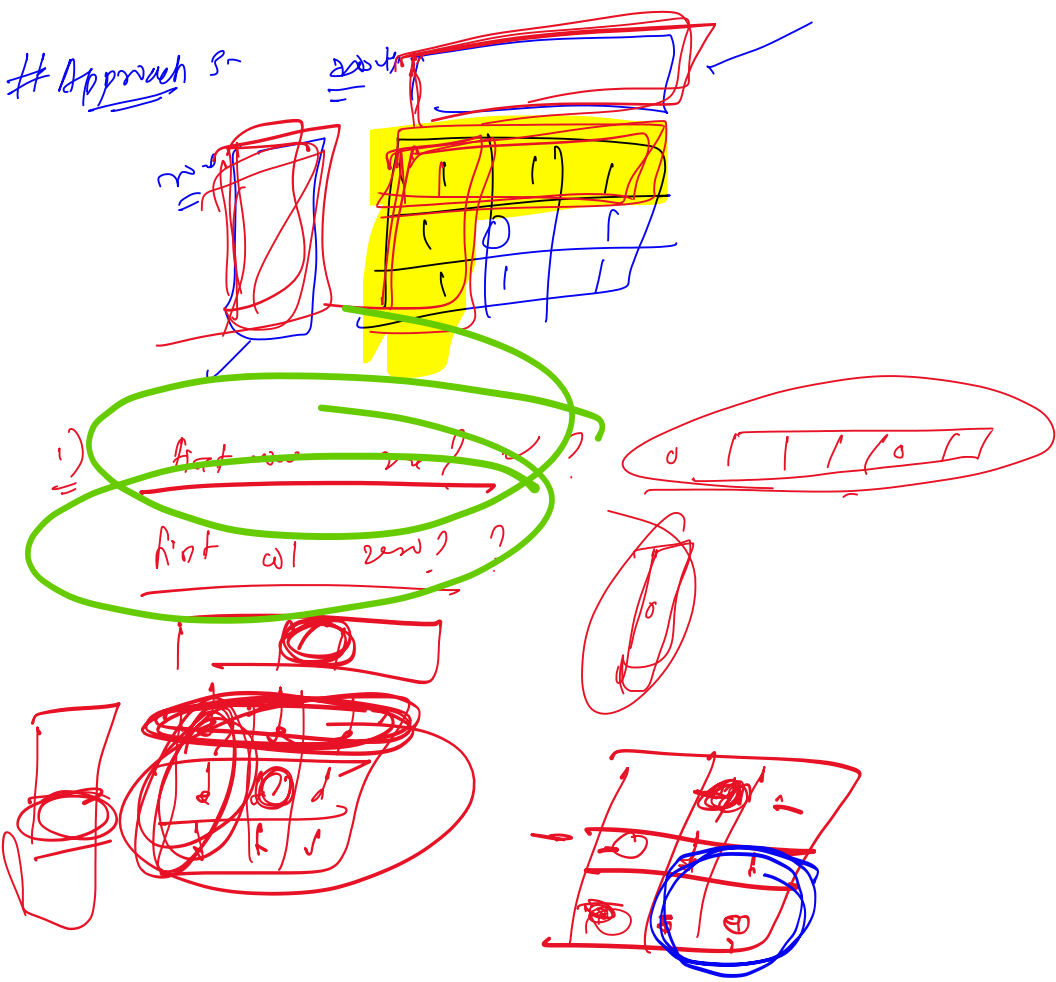




Code-

```
boolean firstRowZero = false;
boolean firstColZero = false;

for (int j=0; j<arr[0].length; j++){
    if (arr[0][j] == 0){
        firstRowZero = true;
        break;
    }
}

for (int i=0; i<arr.length; i++){
    if (arr[i][0] == 0){
        firstColZero = true;
        break;
    }
}
```

```
for (int i=1; i<arr.length; i++){
    for (int j=1; j<arr[0].length; j++){
        if (arr[i][j] == 0){
            arr[i][0] = 0;
            arr[0][j] = 0;
        }
    }
}
```

```
for (int i=1; i<arr.length; i++){
    for (int j=1; j<arr[0].length; j++){
        if (arr[i][0] == 0 || arr[0][j] == 0){
            arr[i][j] = 0;
        }
    }
}
```

```
if (firstRowZero){
    for (int j=1; j<arr[0].length; j++){
        arr[0][j] = 0;
    }
}

if (firstColZero){
    for (int i=1; i<arr.length; i++){
        arr[i][0] = 0;
    }
}
```

$O(m^2n + mn^2)$ $\int O(1)$

$\downarrow$

$O(m \times n)$ $\int O(m+n)$

$\downarrow$

$O(m+n)$ $\int O(1)$